

**Question 13**

The transformation  $T: R^2 \rightarrow R^2$  that maps the graph of  $y = \cos(x)$  onto the graph of  $y = \cos(2x + 4)$  is

A.  $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix}\right)$

B.  $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix}$

C.  $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix}\right)$

D.  $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}\right)$

E.  $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

**Question 14**

The random variable  $X$  is normally distributed.

The mean of  $X$  is twice the standard deviation of  $X$ .

If  $\Pr(X > 5.2) = 0.9$ , then the standard deviation of  $X$  is closest to

A. 7.238

B. 14.476

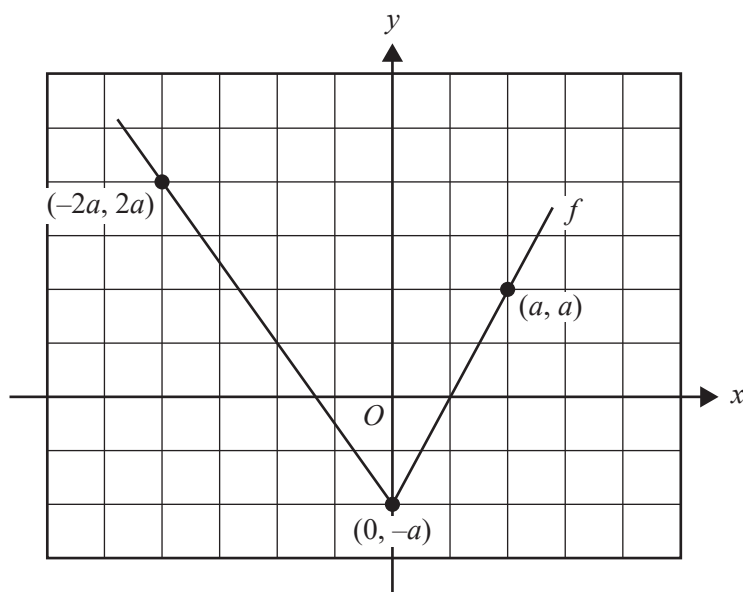
C. 3.327

D. 1.585

E. 3.169

**Question 15**

Part of the graph of a function  $f$ , where  $a > 0$ , is shown below.

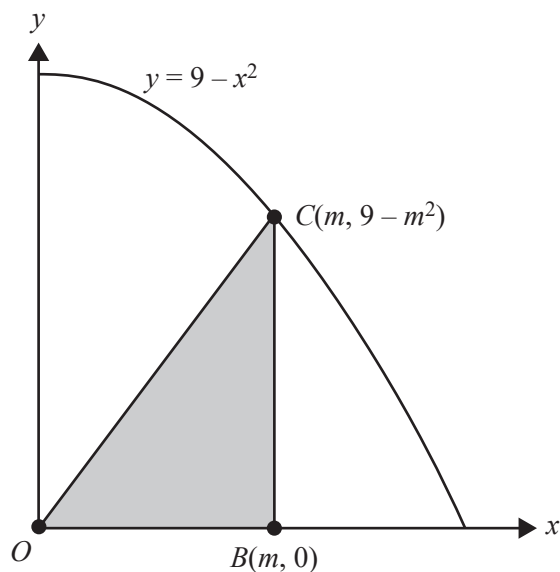


The average value of the function  $f$  over the interval  $[-2a, a]$  is

- A. 0
- B.  $\frac{a}{3}$
- C.  $\frac{a}{2}$
- D.  $\frac{3a}{4}$
- E.  $a$

**Question 16**

A right-angled triangle,  $OBC$ , is formed using the horizontal axis and the point  $C(m, 9 - m^2)$ , where  $m \in (0, 3)$ , on the parabola  $y = 9 - x^2$ , as shown below.



The maximum area of the triangle  $OBC$  is

- A.  $\frac{\sqrt{3}}{3}$
- B.  $\frac{2\sqrt{3}}{3}$
- C.  $\sqrt{3}$
- D.  $3\sqrt{3}$
- E.  $9\sqrt{3}$

**Question 17**

Let  $f(x) = -\log_e(x + 2)$ .

A tangent to the graph of  $f$  has a vertical axis intercept at  $(0, c)$ .

The maximum value of  $c$  is

- A.  $-1$
- B.  $-1 + \log_e(2)$
- C.  $-\log_e(2)$
- D.  $-1 - \log_e(2)$
- E.  $\log_e(2)$

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**Question 18**

Let  $a \in (0, \infty)$  and  $b \in \mathbb{R}$ .

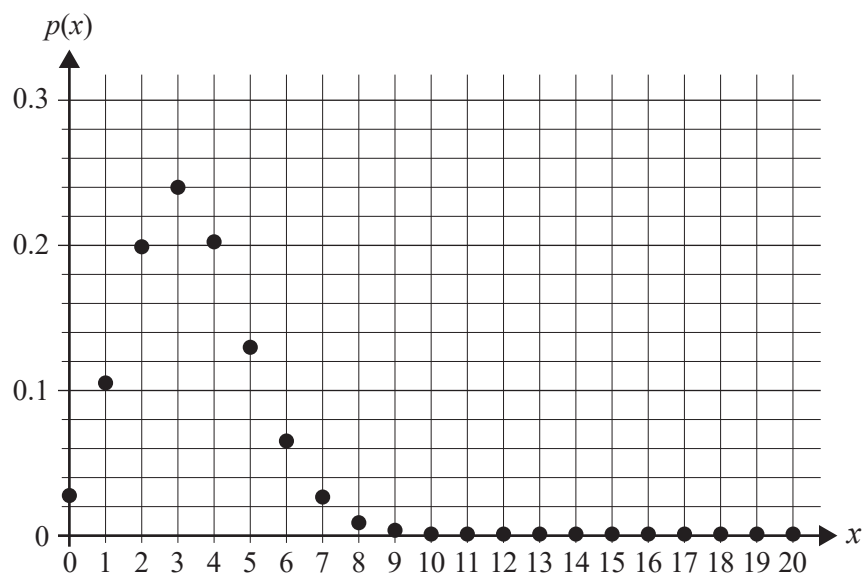
Consider the function  $h: [-a, 0) \cup (0, a] \rightarrow \mathbb{R}$ ,  $h(x) = \frac{a}{x} + b$ .

The range of  $h$  is

- A.  $[b - 1, b + 1]$
- B.  $(b - 1, b + 1)$
- C.  $(-\infty, b - 1) \cup (b + 1, \infty)$
- D.  $(-\infty, b - 1] \cup [b + 1, \infty)$
- E.  $[b - 1, \infty)$

**Question 19**

Shown below is the graph of  $p$ , which is the probability function for the number of times,  $x$ , that a '6' is rolled on a fair six-sided die in 20 trials.



Let  $q$  be the probability function for the number of times,  $w$ , that a '6' is **not** rolled on a fair six-sided die in 20 trials.  $q(w)$  is given by

- A.  $p(20 - w)$
- B.  $p\left(1 - \frac{w}{20}\right)$
- C.  $p\left(\frac{w}{20}\right)$
- D.  $p(w - 20)$
- E.  $1 - p(w)$

**Question 20**

Let  $f: R \rightarrow R$ ,  $f(x) = \cos(ax)$ , where  $a \in R \setminus \{0\}$ , be a function with the property

$$f(x) = f(x + h), \text{ for all } h \in Z$$

Let  $g: D \rightarrow R$ ,  $g(x) = \log_2(f(x))$  be a function where the range of  $g$  is  $[-1, 0]$ .

A possible interval for  $D$  is

- A.  $\left[\frac{1}{4}, \frac{5}{12}\right]$
- B.  $\left[1, \frac{7}{6}\right]$
- C.  $\left[\frac{5}{3}, 2\right]$
- D.  $\left[-\frac{1}{3}, 0\right]$
- E.  $\left[-\frac{1}{12}, \frac{1}{4}\right]$

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**END OF SECTION A  
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## SECTION B

## Instructions for Section B

Answer **all** questions in the spaces provided.

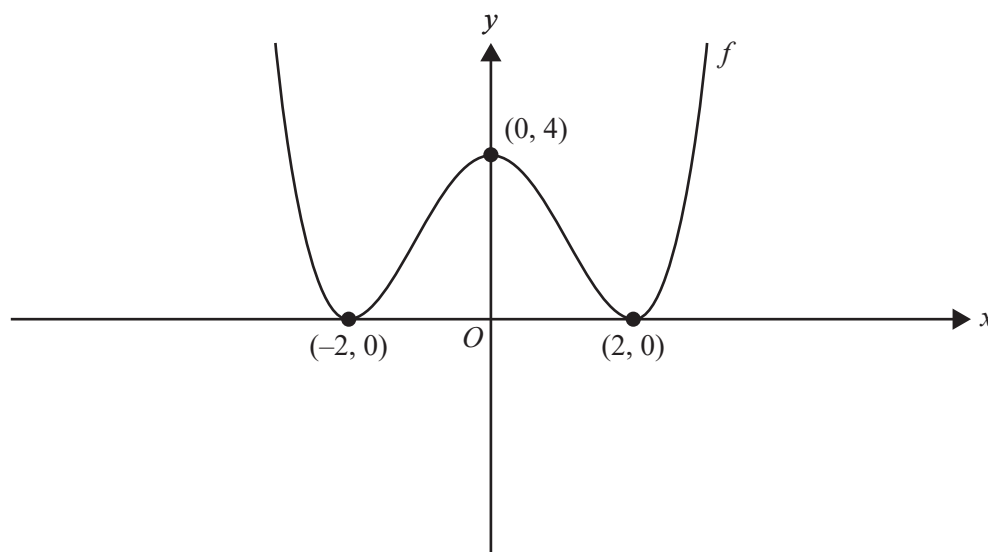
In all questions where a numerical answer is required, an exact value must be given unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

## Question 1 (11 marks)

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = a(x+2)^2(x-2)^2$ , where  $a \in \mathbb{R}$ . Part of the graph of  $f$  is shown below.



- a. Show that  $a = \frac{1}{4}$ .

1 mark

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- b. Express  $f(x) = \frac{1}{4}(x+2)^2(x-2)^2$  in the form  $f(x) = \frac{1}{4}x^4 + bx^2 + c$ , where  $b$  and  $c$  are integers.

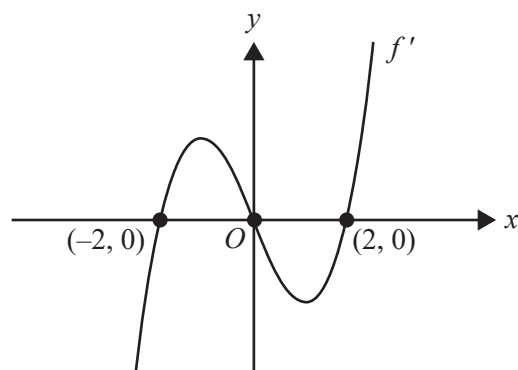
1 mark

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Part of the graph of the derivative function  $f'$  is shown below.



- c. i. Write the rule for  $f'$  in terms of  $x$ . 1 mark

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- ii. Find the minimum value of the graph of  $f'$  on the interval  $x \in (0, 2)$ . 2 marks

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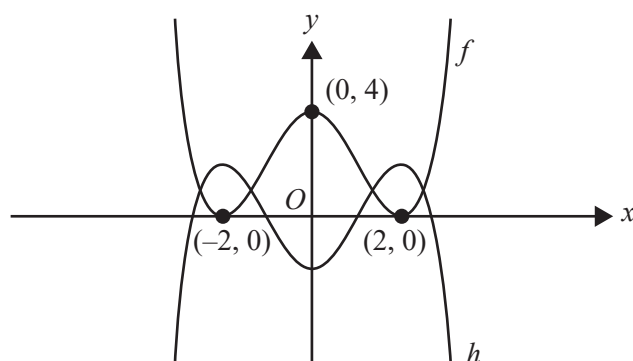


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Let  $h: \mathbb{R} \rightarrow \mathbb{R}$ ,  $h(x) = -\frac{1}{4}(x+2)^2(x-2)^2 + 2$ . Parts of the graphs of  $f$  and  $h$  are shown below.



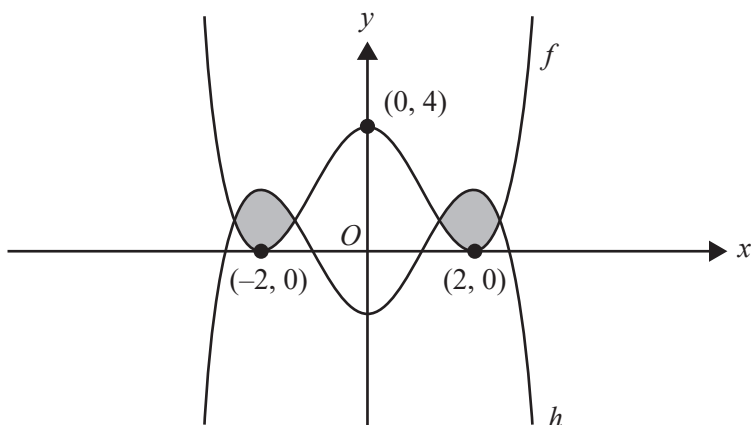
- d. Write a sequence of two transformations that map the graph of  $f$  onto the graph of  $h$ . 1 mark

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- e. i. State the values of  $x$  for which the graphs of  $f$  and  $h$  intersect. 1 mark

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- ii. Write down a definite integral that will give the total area of the shaded regions in the graph above. 1 mark

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- iii. Find the total area of the shaded regions in the graph above. Give your answer correct to two decimal places. 1 mark

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- f. Let  $D$  be the vertical distance between the graphs of  $f$  and  $h$ .  
Find all values of  $x$  for which  $D$  is at most 2 units. Give your answers correct to two decimal places. 2 marks

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**SECTION B – continued**  
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**Question 2** (11 marks)

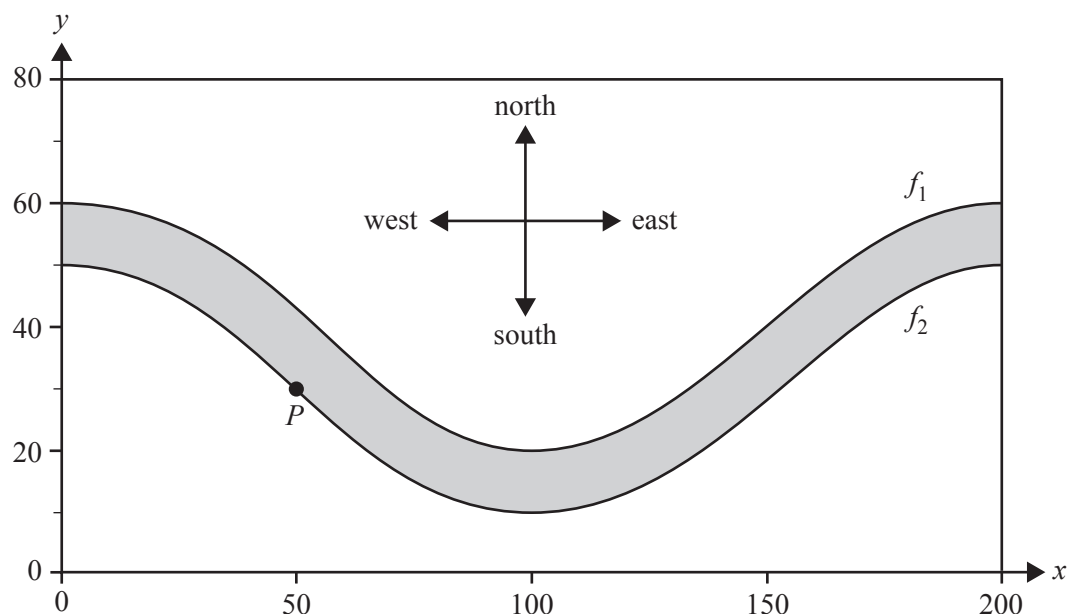
An area of parkland has a river running through it, as shown below. The river is shown shaded.

The north bank of the river is modelled by the function  $f_1 : [0, 200] \rightarrow \mathbb{R}$ ,  $f_1(x) = 20 \cos\left(\frac{\pi x}{100}\right) + 40$ .

The south bank of the river is modelled by the function  $f_2 : [0, 200] \rightarrow \mathbb{R}$ ,  $f_2(x) = 20 \cos\left(\frac{\pi x}{100}\right) + 30$ .

The horizontal axis points east and the vertical axis points north.

All distances are measured in metres.



A swimmer always starts at point  $P$ , which has coordinates  $(50, 30)$ .

Assume that no movement of water in the river affects the motion or path of the swimmer, which is always a straight line.

- a.** The swimmer swims north from point  $P$ .

Find the distance, in metres, that the swimmer needs to swim to get to the north bank of the river.

1 mark

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